

MACQUARIE ISLAND, Australia

WMO: 949980

Lat: 54.499S

Lon: 158.937E

Elev: 27

StdP: 14.68

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.8	29.0	17.7	13.5	28.6	20.0	15.0	30.4	47.7	40.5	44.7	40.3	19.3	190	1.217

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.7	49.4	47.5	48.6	46.8	48.0	46.3	48.4	48.8	47.6	48.1	46.8	47.4	22.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
48.1	49.7	48.5	47.2	48.0	47.8	46.4	46.6	47.2	19.5	48.9	19.1	48.3	18.7	47.7	53.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.6	41.0	37.7	DB	23.5	51.6	3.3	1.5	21.1	52.7	19.2	53.6	17.3	54.4	14.9	55.5	(4)
(5)			WB	21.6	50.0	2.9	1.1	19.5	50.8	17.8	51.5	16.2	52.1	14.1	53.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	44.4	44.7	43.6	42.2	40.1	38.4	37.8	38.3	38.4	39.1	40.2	43.0
	DBStd	3.81	2.02	2.03	2.20	2.53	3.28	3.71	3.76	3.14	3.29	2.72	2.83	2.50
	HDD50	3350	173	149	199	235	307	348	380	364	348	337	294	216
	HDD65	8825	638	569	664	685	772	798	845	829	798	802	744	681
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	22.7	21.2	22.5	23.5	24.4	23.5	23.1	22.9	22.7	23.1	23.6	21.9	20.3
Precipitation	PrecAvg	37.5	3.3	3.3	3.8	3.7	3.4	3.1	2.9	2.9	2.9	3.1	2.9	3.1
	PrecMax	48.9	6.9	5.1	6.5	6.9	5.7	6.1	5.0	6.1	5.3	5.0	5.4	7.0
	PrecMin	28.2	1.2	0.9	1.2	2.0	2.0	1.2	0.9	1.1	1.0	1.5	1.3	1.3
	PrecStd	5.0	1.1	0.9	1.1	1.0	0.9	1.0	1.0	1.1	0.9	0.9	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	50.8	49.4	48.4	47.1	45.5	45.0	44.8	45.0	45.9	47.4	50.4
		MCWB	47.7	48.9	47.8	47.0	46.1	43.9	43.7	43.4	43.4	44.1	45.0	48.5
	2%	DB	49.2	49.4	48.4	47.5	45.8	44.6	44.0	43.9	44.1	44.7	45.9	48.8
		MCWB	46.9	47.7	47.0	46.2	44.7	43.2	42.9	42.7	42.7	43.1	43.9	47.0
	5%	DB	48.4	48.5	47.6	46.7	45.0	43.9	43.2	43.2	43.6	43.9	45.0	47.6
		MCWB	46.3	46.8	46.3	45.5	43.9	42.7	42.1	42.1	42.3	42.4	43.2	45.9
	10%	DB	47.6	47.7	46.9	45.9	44.3	43.2	42.5	42.6	42.9	43.2	44.1	46.7
		MCWB	45.7	46.1	45.5	44.7	43.2	42.2	41.5	41.6	41.8	41.8	42.5	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	49.8	48.7	47.6	46.6	44.5	44.1	44.0	44.1	45.0	45.9	49.4
		MCDB	49.7	50.3	48.9	47.8	46.8	44.8	44.5	44.2	44.3	45.3	46.5	49.9
	2%	WB	47.9	48.4	47.7	46.7	45.0	43.7	43.2	43.2	43.4	43.8	44.7	47.6
		MCDB	48.4	48.8	48.0	47.1	45.4	44.1	43.6	43.5	43.8	44.3	45.4	48.3
	5%	WB	47.0	47.5	46.8	45.9	44.3	43.1	42.5	42.5	42.7	43.0	43.8	46.5
		MCDB	47.7	48.0	47.2	46.5	44.7	43.6	43.0	42.9	43.3	43.7	44.6	47.2
	10%	WB	46.2	46.7	45.9	45.0	43.6	42.4	41.8	41.8	42.0	42.2	43.0	45.5
		MCDB	47.1	47.4	46.6	45.7	44.1	43.1	42.4	42.5	42.7	43.1	43.9	46.3
Mean Daily Temperature Range	5% DB	MDBR	5.0	4.7	4.6	4.4	4.8	5.0	4.9	4.9	5.3	5.5	5.3	5.2
		MCDBR	5.8	5.1	4.9	4.4	4.9	4.6	4.9	4.9	4.9	5.3	5.5	5.7
		MCWBR	5.2	5.0	5.5	5.3	6.0	5.4	5.7	5.7	5.4	5.8	5.2	5.0
	5% WB	MCDBR	5.2	4.6	4.8	4.5	4.8	4.6	4.9	4.9	4.9	5.1	5.0	5.2
		MCWBR	5.3	5.0	5.6	5.5	5.8	5.5	5.7	5.7	5.4	5.8	5.1	4.9
Clear-Sky Solar Irradiance	taub	0.333	0.331	0.329	0.319	0.315	0.298	0.299	0.321	0.338	0.333	0.329	0.328	
	taud	2.535	2.551	2.542	2.510	2.316	2.265	2.362	2.444	2.440	2.488	2.520	2.549	
	Ebn at Noon	307	295	274	240	187	165	193	233	265	291	306	312	
	Edh at Noon	32	29	25	20	17	14	16	21	28	30	32	32	
All-Sky Solar Radiation	RadAvg	1588	1252	796	412	198	118	158	330	666	1090	1470	1660	
	RadStd	98	62	59	29	17	9	13	27	36	86	96	129	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.45	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page